Marking various aspects in Turkish Sign Language

BİT ('finish') and 'bn'

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Sign languages have been reported to have manual signs that function as perfective morphemes (Fischer & Gough 1999; Meir 1999; Rathmann 2005; Duffy 2007; Zucchi et al. 2010). Turkish Sign Language (TİD) has also been claimed to have such morphemes (Zeshan 2003; Kubuş & Rathmann 2009; Dikyuva 2011; Gökgöz 2011; Karabüklü 2016) as well as a nonmanual *completive* marker ('bn') (Dikyuva 2011). This study shows that the nonmanual 'bn' is in fact a perfective morpheme. We examine its compatibility with different event types and furthermore show that TİD has a manual sign BİT ('finish') that is indeed the completive marker but with possibly unusual restrictions on its use. Based on their distribution, the current study distinguishes BİT and 'bn' as different morphemes even though they can co-occur. TİD is argued to be typologically different from other sign languages since it has both a nonmanual marker ('bn') for a perfective morpheme and a manual sign (BİT) with different selectional properties than the manual signs reported for other sign languages.

Keywords: Turkish Sign Language (TİD), aspect morphemes, perfectivity, event types, nonmanual markers

1. Introduction

This paper focuses specifically on two TİD morphemes which have been analyzed as aspect morphemes of various types. The goal is to determine what separates these two morphemes – the manual morpheme BİT ('finish') (Figure 1) and the nonmanual morpheme 'bn'[1] (tongue protruding) (Figure 2). Based on their com-

1. The label 'bn' was first used by Dikyuva (2011: 25) to refer to this specific nonmanual marker. His convention will be used in this paper to have a consistent reference in the literature.

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patibility with different event types, we argue that 'bn' is a perfective morpheme that bounds the event. In contrast, BİT is a completive morpheme that occurs with Ramchand's (2008) event category [init, path], some events in the category [proc], and with at least some statives. These findings show that TİD is typologically different from other sign languages by having a nonmanual marker ('bn') for a perfective morpheme and having a manual morpheme (BİT) for a completive morpheme.

TELL BİT

'i (have) told (it).'

Figure 1. Manual sign BİT ('finish')

bn,pq

READ

'Did/Have you read (it)?'

Figure 2. Nonmanual morpheme 'bn'[2]

As seen in sentence (1), depicted in Figure 1, BİT[3] occurs after the main verb TELL and indicates that the event of telling is completed. In the literature, BİT has been referred to as an aspect marker (Kubuş & Rathmann 2009; Gökgöz 2011)

2. The polar question nonmanual marker in this sentence is presented as 'pq' next to 'bn' because both have the same spreading domain over the single sign READ.

3. In general, sign language glosses in this study are presented in English. However, aspectual signs are given in previous authors' glossing to make clear which sign language is under discussion, e.g., FATTO ('done') from Italian Sign Language (Zucchi et al. 2010) and BİTTİ ('done') from TİD (Zeshan 2003).

and a completive marker (Zeshan 2003). We will present an analysis to support BİT's behaviour as a completive marker based on its acceptability with event types.

Researchers also noted that TİD may have other potential manual aspect markers such as BİTTİ ('done'), TAMAM ('done') (Zeshan 2003), and SON ('end') (Dikyuva 2011). Yet, the aspectual features of BİT ('finish'), BİTTİ ('done'), TAMAM ('done') and SON ('end') have not been fully analyzed in terms of compatibility with different event types. The current study defines the aspectual properties of only BİT ('finish'), leaving the other morphemes for future studies due to time constraints.

(1) The signer was instructed to imagine that she is giving a TİD class and that she was going to start a new topic, and to tell the class that she explained/covered ('told') the previous chapter.
IX$_1$ TELL BİT
'I (have) told (it).'

In the absence of attention to the selectional properties of BİT in terms of event types, it is clear why it was considered to be the dedicated manual sign for perfective marking, similar to aspectual signs occurring in other sign languages, such as FINISH in American Sign Language (ASL; Fischer & Gough 1999; Rathmann 2005; Duffy 2007; Zucchi et al. 2010), FATTO ('done') in Italian Sign Language (LIS; Zucchi et al. 2010), and ALREADY in Israeli Sign Language (ISL; Meir 1999; Meir & Sandler 2008). However, we will show that BİT is acceptable with stative events as well as with events in [init, path] and some events in [proc] categories, in contrast to perfective manual signs in other sign languages, which were reported to be acceptable with all well-known event categories – accomplishment, achievement, and activity verbs – but not state verbs.

As for 'bn', it occurs over the main verb as in (2a) and, like BİT, it also shows that the event of telling has ended. Dikyuva (2011) analyzed 'bn' as a completive marker in TİD based on its compatibility with action, process and state verbs. In contrast, Kubuş & Rathmann (2009) analyzed 'bn' as a [past, telic] marker by reporting that it is only compatible with activity verbs and not with other event types. Thus, these contradictory findings necessitate re-examination of 'bn' with different event types to better understand its aspectual features.

(2) The signer was instructed to imagine that she is giving a TİD class and that she was going to start a new topic, and to tell the class that she explained/ covered ('told') the previous chapter.

a. IX$_1$ $\overline{\text{TELL}}^{\text{bn}}$
'I (have) told (it).'

b. $\overline{\text{bn}}$
IX$_1$ TELL BİT
'I (have) told (it).'

Even though they have been separately reported to be aspectual morphemes, we see that these two morphemes can co-occur[4] as in sentence (2b). This situation also forces us to have another look at the functions of BİT and 'bn'.

After presenting the proposals made in previous studies regarding BİT and 'bn' in Section 2, Section 3 presents the methodology used to answer the research questions. A general overview of BİT and 'bn' is given in Section 4 to show that they cannot be freely substituted for each other. Section 5 presents their temporal features to show that they are not tense markers. Section 6 demonstrates that their negative counterparts are different from sentential negation, and that the morphemes BİT and 'bn' follow the general trend in sign language typology. Section 7 presents their compatibility with different event types following Ramchand's (2008) classification and shows clearly that BİT and 'bn' have different selectional properties. BİT is acceptable with statives, [init, path], and some [proc] verbs while 'bn' is acceptable with all eventive categories and stative verbs. Section 8 concludes.

2. Previous studies on BİT and 'bn'

BİT has been mentioned to be an aspect marker (Gökgöz 2011) and a completive marker (Zeshan 2003), among other functions (see Section 4). BİT in sentence (3) was reported in Gökgöz (2011: 54) as an aspectual marker conveying the completion of the event. Yet, further justification for BİT being an aspectual sign in sentence (3) was not provided in either Zeshan (2003) or Gökgöz (2011). It is crucial to know whether BİT is acceptable with all event types like the manual aspect signs in other sign languages. Furthermore, such acceptability patterns should also show whether BİT and 'bn' are different realizations of the same concept.

(3) IDEA CONFUSED WRITE BİT[5] (TİD, adapted from Gökgöz 2011: 55)
'He finished writing his confused ideas.'

4. Even though their co-occurrence needs further in-depth analysis, initial findings are presented in Appendix I.

5. The examples cited in this paper are given in English glosses except for the aspect markers, even for those examples that are glossed in both Turkish and English in the original sources.

As for 'bn', Dikyuva (2011: 47) analyzes it as a completive marker based on the explanation that "[...] 'bn' occurs with most of action, process and state[6] verbs". On the other hand, Kubuş & Rathmann (2009) report that 'bn' is unacceptable with states, semelfactives, achievements, and some accomplishments. They analyze 'bn' as a [past, telic] marker based on its compatibility with activities and incompatibility with future time adverbials. Since they report 'bn' as not acceptable with all event types, they argue that it is not a perfective morpheme.[7] Although both studies analyze 'bn' similarly as completive or [past, telic] marker, Kubuş & Rathmann (2009) claim it is acceptable only with activities while Dikyuva (2011) claims that it is acceptable also with states and processes. This forces reexamination of 'bn' systematically with more verbs from the event classes.

Due to the insufficiently detailed analysis of BİT and the contradictory findings reported for 'bn', a detailed analysis with event types is needed to answer whether BİT and 'bn' are the realizations of the same notion. In other words, the research question here is whether BİT and 'bn' are the manual and nonmanual markers realizing the same aspectual category, or if they have different semantic functions.

3. Data sources

This section presents an overview of the data employed to describe the semantic features of BİT and 'bn'. Since our aim was to understand the basics of BİT and 'bn' and their compatibility with event types, time adverbials and negation, a study designed as an experiment that would generate descriptive statistics and statistical significance (of participant variables, acceptability ratings) is premature. We first need to have the basics in order to design such an experiment. The data used here comes from three different sources: (i) existing data collected for two previously funded projects, (ii) elicited data that we collected in two ways: semantic elicitation tasks and judgment tasks.

6. Dikyuva (2011) did not discuss whether he makes a distinction between statives and states; he only uses the term 'states'.

7. Additionally, Kubuş & Rathmann (2009) argue against it being a perfective morpheme because 'bn' does not allow narrative advancement. However, languages may grammaticalize the different functions of semantic notions in one morpheme. It is not clear that narrative advancement behaves as a uniform test for perfectivity in this regard (Dahl 1985). Similarly, the judgments cited here also highlight that perfective context is not enough to license 'bn', and that it needs further support in the discourse. Nonetheless it behaves as a perfective marker whenever it is licensed in the discourse; the licensing motivations will be left for future studies.

3.1 Existing data / corpus data

Here, by 'corpus' we mean 'existing body of data, recordings' that were collected for research purposes. The existing data come from the recordings of two funded projects: (1) COST Action IS1006, 'Unraveling the Grammars of European Sign Languages: Pathways to Full Citizenship of Deaf Signers and to the Protection of their Linguistic Heritage' (2012–2015), and (2) Machine Translation and Avatar Animation from Turkish to Turkish Sign Language (TÜBİTAK project no. 114E263 2014–2016). Some of the recordings for these projects were glossed but did not have nonmanual marker annotations. Thus, we were able to search the glosses for only BİT and not for 'bn'. We categorized the 400 instances of BİT into lexical verb, discourse marker, and functional usage (Table 1).

Table 1. Number of instances of BİT in glossed annotations

Functions	Number of instances
Lexical verb	200
Discourse marker	170
Functional usage	130
Total	**400**

The category 'lexical verb' includes the instances where BİT was the only verb, preceded by or followed by an adverb, or followed by sentential negation. The category 'discourse marker' includes instances where BİT appeared as the last sign of the given production to signal to the interlocutor that the turn was completed. The last category 'functional usage' includes the instances where BİT appeared after another verb, or after a sequence of verbs, or after a clause but not the end of the main sentence. All instances that could not be clearly classified were discussed with one of the team's deaf members.

3.2 Elicited data: Production and judgment tasks

3.2.1 *Participants*

Sixteen signers (11 f, 5 m) aged 24–63 participated in this study. All of the participants were active members of the deaf community and were living in Istanbul at the time of data collection. The language profile of the participants was mixed: three deaf-of-deaf, the rest self-reported fluent signers. The same participants took part in all tasks.

3.2.2 *Materials*

For the semantic production elicitation, picture sequences were taken from Totem Field Storyboards that were specifically designed for semantic fieldwork studies. The storyboards used were Chameleon Story (Group 2012), Bill vs The Weather (Vander Klok 2013), and Thank-you Notes (Littell 2010). Figure 3 presents an excerpt from the Chameleon Story.

Figure 3. An excerpt from the picture sequence 'Chameleon Story'

For semantic judgment tasks, contexts that entail different event types and target sentences for the contexts were created based on previous findings. The contexts were discussed with the deaf team member for their naturalness. In addition to the contexts and their target sentences, sentences with different time adverbials, sentential negation, and negative forms of BİT and 'bn' were recorded with one deaf member to be used as stimuli.

3.3.3 *Procedure*

The participants were presented the tasks in randomized order. All tasks were conducted by one of the team's deaf members at that time.

As for the semantic elicitation tasks, the participant was randomly given one of the picture sequences. They were asked first to look at the pictures, then sign the story to the deaf assistant. The assistant followed the picture sequence.

As for the semantic judgment tasks, the deaf team member first signed the context to the participant, and the participant was asked how to respond. Then, the target sentence containing the aspect morpheme BİT or 'bn' was signed, and the participant was asked whether the sentence was acceptable in the context. The participant could ask any questions about the context or the target question. If the sentence was deemed unacceptable, the participant was asked how to sign it to fit the context.

Lastly, for the target sentences with time adverbials, negation, and negative forms, the participant was asked to watch the pre-recorded videos, and was then asked whether the sentence was acceptable. If the sentence was unacceptable, the participant was requested to sign it in an acceptable form.

3.4 Summary

In the remainder of the paper, the discussion of BİT is based on both corpus and elicited data, whereas the discussion of 'bn', as previously mentioned, is necessarily restricted to only the elicited data. The influences of context, different event types, time adverbials, and negation were analyzed. The following sections lay out the temporal and aspectual features of BİT and 'bn'.

4. General overview of BİT and 'bn'

The main focus here is to describe the aspectual features of BİT and 'bn'. We begin with the various semantic or pragmatic contexts in which they appear. BİT appears at the end of a sentence, discourse or narration to show that an event or task in the discourse or narration has come to an end. In contrast, 'bn' appears in past, perfect and impersonal sentences. Thus, they cannot be used interchangeably in every sentence, and this supports the suggestion that they are not the realization of the same notion. Below, this is laid out in more detail for each morpheme.

4.1 Different functions of BİT

Consistent with Zeshan's (2003) findings on BİT, the findings from the TİD corpus show that BİT has various functions: a main verb, an aspect marker, a marker sequencing events, or a discourse marker. For this report, instances of BİT occurring in sentences from the corpus were further discussed with TİD signers for their meaning and their acceptability with aspect interpretation. Since TİD is a pro-drop language (Sevinç 2006), in (4) (Figure 4), BİT is the main verb of the second clause (reflected by the comma notation) and refers to the completion of the game rather than the completion of the event PUT.

(4) Context: The participants were working on a lego task. One was giving instructions to the other participant to build the lego structure. The participant gave the last instruction in the following sentence and signed that they were done.
PURPLE PUT, BİT[8] (main verb)
'You put the purple (brick), (the game) finishes.'

8. Commas are commonly used in sign language glossing whenever there is attention to the syntactic structure containing more than one clause, or to the separation of the topic from the main clause, especially when there is a prosodic separation. We also use it in the paper to indicate when BİT and the verb are not part of the same clause.

PURPLE PUT BİT

'You put the purple (brick), (the game) finishes.'

Figure 4. BİT as a main verb

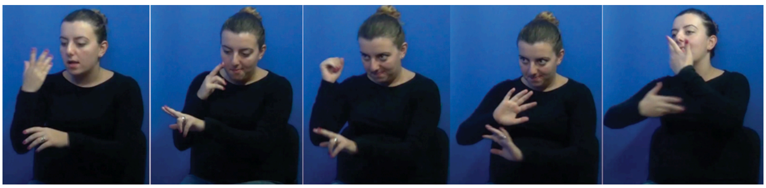
Furthermore, when BİT is meant as an aspect marker in the same clause as PUT in a variant of (4), it is not acceptable (5) (see Section 7.2 for details). As an aspect marker, there are verbs that BİT can co-occur with like DRINK, as reported in previous studies (Gökgöz 2011; Kelepir et al. 2019; Zeshan 2003). For example, BİT can function as an aspect marker in sentence (6), where it is not the main verb, but rather conveys that the event of drinking has been completed. Whether BİT is functioning as a main verb or an aspect marker in a given sentence can be tested with the negated counterpart of that sentence, as the aspect marker BİT is not negated by means of the standard sentential negation strategy (see Section 6 for details).

(5) #PURPLE PUT BİT (aspect marker)
'You put/have put the purple (brick).'

(6) Context: The participants were role-playing as doctor-patient. The one who was the patient was explaining to the doctor what s/he ate, drank all day.
TURKISH COFFEE DRINK BİT (aspect marker)
'I drank/ have drunk Turkish coffee.'

Being a main verb and an aspect marker are not the only functions of BİT. The corpus contains examples in which BİT sequences events (7) (Figure 5) or signals the end of the discourse (8) (Figure 6). BİT in (7) sequences two events ('looking and memorizing' and 'telling') and behaves like an adverbial such as 'after' or 'when'. Note that despite its appearance next to the sign UNDERSTAND, it does not modify UNDERSTAND, which is a change-of-state verb, but rather sequences the event of which UNDERSTAND is part (see Section 7.2 for detailed discussion). We also observe the same reading ('you are done') in sentence (8). The signer gives a recipe instruction, and she uses BİT to convey that the interlocutor is done; it is the end of the recipe. BİT in (8) occurs as the last sign of the discourse, and the signer completes her instructions with BİT. Like sentence (7), BİT does not modify the verb POUR, and it is not part of the previous clause (CL-TRAY POUR).

(7) Context: Deaf consultant gives instructions to the deaf participants on a task. One of the participants is asked to watch a video and tell what s/he watched to the other participant.
WATCH LOOK$_3$ MEMORIZE BİT, $_2$TELL$_1$ (sequencing events)
'After you watch and memorize (it), tell it to me.'

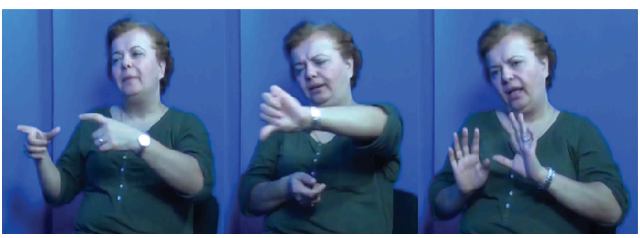


WATCH LOOK$_3$ MEMORIZE BİT $_2$TELL$_1$

'After you watch and memorize (it), tell it to me.'

Figure 5. BİT used for sequencing events

(8) Context: Deaf participant gives a recipe to another participant and she signs that it is the end of the recipe.
CL-TRAY POUR, BİT (discourse marker)
'Pour (it) into the tray, and you are done.'



CL-TRAY POUR BİT

'Pour (it) into the tray and you are done.'

Figure 6. BİT used as a discourse marker

In addition to these examples, BİT is observed to convey various meanings which only become clear in the discourse. The discussants highlighted that these sentences are acceptable only in certain contexts and that they do not just convey the termination or the completion of the event. For instance, the sentences in (9) and (11) are acceptable in the given contexts, whereas sentences (10) and (12) do not convey the aspectual interpretation when there is no context. Similar to previous Examples (7) and (8), these sentences also convey that the signer is done with the event (9) or the responsibility (11). Sentences (9) and (12) involve the events ‘drive’ or ‘grow’ as processes that can be considered as ongoing and uncompleted. However, the signers in these contexts convey that the respective event is terminated. In other words, the signers counteract the expectation of incompleteness by using BİT. As a discourse marker, BİT might be playing a role regarding the at-issueness of event completion, but this requires a detailed future study on the pragmatics of BİT.[9]

(9) Context: The signer is taking the driving test and his/her friend is, too. To check in with the signer, his/her friend asks whether s/he has taken the test or not. The signer replies:
DRIVE BİT
‘I have driven the car, and my turn is over. (I am done with driving.)’

(10) Context: *Out of the blue*/*no specific context*
#DRIVE BİT
‘I have driven.’

(11) Context: The signer’s friend has not seen him/her for a while. When they see each other and chat, the friend asks about the signer’s kid.
CHILD GROW BİT IX_1
‘The child has grown up, and my responsibility is over.’

(12) Context: *Out of the blue*/*no specific context*
#CHILD GROW BİT
‘The child has grown.’

These findings from the TİD corpus support the claims of Zeshan (2003). BİT has various functions: main verb (4), aspect marker (6), sequencing events (7), and discourse marker (8). Similar to what has also been reported for FINISH in ASL (Fischer & Gough 1999), the various functions of BİT seem not to be unrelated and might share a root in one core semantic function. The interesting finding is that as an aspect marker, BİT is compatible with certain verbs like DRINK. To lay out its difference from ‘bn’, this study will focus on the aspectual properties of BİT in Section 7.2. Here we continue to separate it from ‘bn’.

9. We want to thank the reviewer who suggested this possible explanation for the behavior of BİT.

4.2 Various structures in which 'bn' occurs

The nonmanual marker 'bn' appears in different sentence structures such as past (Kubuş & Rathmann 2009), perfect, or impersonal (Kelepir & Özkul 2015; Kelepir et al. 2019). Unlike BİT, 'bn' has the same function in all these sentences, that is, the event is completed or terminated. First, 'bn' can occur with past tense adverbials (13). The verb occurs with 'bn'; time is conveyed with the adverbial LAST WEEK since TİD is a morphologically tenseless language (Karabüklü 2018). Yet, the same sentence with BİT (14) is unacceptable due to the event type of ARREST (see Section 7.2 for details).

(13) LAST WEEK THIEF $\overline{\text{ARREST}}^{\text{bn}}$
'They arrested the thief last week.' (Kelepir & Özkul 2015, slide 24)

(14) #LAST WEEK THIEF ARREST BİT
'They arrested the thief last week.' (Kelepir & Özkul 2015, slide 24)

Also, it is possible to delete the time adverbial LAST WEEK from sentence (13), and the sentence is still acceptable. In contrast to sentence (13), the time of the event is unknown in sentence (15), which only conveys that there exists such an event, that is, the perfect reading. Thus, without a context or an adverbial, 'bn' can occur in past or perfect sentences. It still conveys that the event is actualized. The BİT counterpart of sentence (15) is unacceptable (16). The difference between (14) and (16) shows that the unacceptability is due to the event and not due to the time adverbial.

(15) THIEF $\overline{\text{ARREST}}^{\text{bn}}$
'They have arrested the thief.'

(16) #THIEF ARREST BİT
'They have arrested the thief.'

As reported by Kelepir et al. (2019), 'bn' can also appear in impersonal sentences. In sentence (17), it conveys that the event has been realized and is completed. Its counterpart with BİT is likewise unacceptable (18). Sentence (17) lacks an agent, and this is how the impersonal reading is derived in combinations with nonmanual markers and verb agreement (Kelepir et al. 2019). Hence, the readings in sentences (13), (15) and (17) are derived in the context or from the combination with other elements like adverbials, nonmanuals, or lacking an agent. Even though 'bn' appears in past, perfect and impersonal sentences, the meaning of 'bn' stays constant as the event is completed.

(17) WINDOW $\overline{\text{BREAK}}^{\text{bn}}$
'The window was/has been broken.' (Kelepir & Özkul 2015, slide 24)

(18) #WINDOW BREAK BİT
'The window was/has been broken.' (Kelepir & Özkul 2015, slide 24)

As an interim summary, BİT conveys an end either in a single clause (6) or in a sequence of clauses (8). It conveys a slightly different but related meaning that can be paraphrased as 'come to an end'. In this regard, BİT behaves like FINISH in ASL (Fischer & Gough 1999). In contrast, 'bn' has the same function in all sentences – past (13), perfect (15), or impersonal (17) – in that it conveys that the event has reached its end. Hence, one cannot easily replace BİT and 'bn' for each other in these sentences, and the sentences would not convey the same reading with either morpheme, as seen in the sentence pairs (13)/(14), (15)/(16) and (17)/(18). Thus, the structures that BİT and 'bn' appear in show that they are different morphemes. To further support this claim and to illustrate the difference between the two morphemes, their temporal and aspectual properties are tested in Sections 5 and 7.

5. Temporal features of BİT and 'bn'

The marker 'bn' has been reported to convey [past, telic] by (Kubuş & Rathmann 2009), and it combines with past tense adverbials (14). As for BİT, it had not been tested with time adverbials but one of its function is sequencing events. There is a possibility that it might convey past or anteriority. To test whether BİT and 'bn' have specific temporal meaning or are tense markers, they were investigated in the elicitations with imperfective verb forms and time adverbials denoting past, present and future.

As for theoretical approaches to time and tense, there are two main views: one is to treat it as a variable or index based on the analogies with pronouns (Abusch 1997; Kratzer 1998; Partee 1973), and the other is to treat it as quantificational (von Stechow 1995, 2009). Since the focus of the current study is on the aspectual properties of BİT and 'bn', we will briefly introduce Reichenbach's (1947) definition of tense and leave the detailed analysis for the future. Section 5.1 introduces Reichenbach's (1947) approach to tense, while Section 5.2 presents the tests with BİT and 'bn'.

5.1 Reichenbach's definition of tense

Reichenbach (1947) considers time metaphorically as a line and postulates points on the line. Tense is then the relation among the points. The first two points are Utterance Time (UT) and Event Time (ET). Present is considered as UT. ET is the time when an event is realized. The combinations of these points give three different possibilities, namely ET < UT (past) (19), ET = UT (present) (20), and UT < ET (future) (21). A third point on the time line is Reference Time (RT) that can be the same as UT, as in sentences (19), (20) and (21). RT can be specified in the context or in the sentence via time adverbials. For instance, in (22), UT is 'now' and RT is provided by the phrase 'by May', whereas ET is not specified in the sentence and can be any point between 'now' and 'May'.

(19) The project was completed last May. [ET < UT]

(20) The project continues. [ET = UT]

(21) The project will be completed next April. [UT < ET]

(22) The project will be completed by May. [UT ≤ ET < RT]

The relations among ET, UT, and RT do not address the internal temporal structure of an event but indicate where the event is situated with respect to UT. Based on the description of tense, the prediction is that any event can be presented in these relations as in (23) and (24). The tense of both (23) and (24) is past tense since ET precedes UT. The only difference in these sentences is how the event 'painting a picture' is viewed.

(23) The kid was painting a picture.

(24) The kid painted a picture.

Building upon Reichenbachs's (1947) universal tense definition, BİT and 'bn' were tested with imperfective verb forms because any event form should be able to be presented in any tense form as in (23) and (24). Thus, if they were tense markers, they should both be able to appear with imperfective forms. Moreover, if they convey a specific tense relation, they should not appear with adverbials that convey different times. Hence, BİT and 'bn' were also tested with different time adverbials.

5.2 Testing BİT and 'bn'

Participants were presented with BİT and 'bn' with imperfective verb forms[10] to test whether they are tense markers. As seen in (25a) and (25b), both BİT and 'bn' are unacceptable with imperfective verb forms. Incompatibility eliminates the possibility of them being tense markers based on the assumption that any event form should be able to be represented in any tense form.

(25) Context: You have a child, and you were not home that day. You came home and asked your partner how they spent their day. Your partner told you that the child was playing for the last two hours.

a. *CHILD TWO^HOUR PLAY.IMPF BİT
Intended: 'The child has been playing for two hours. / The child was playing for the last two hours.'

b. *CHILD TWO^HOUR $\overline{\text{PLAY.IMPF}}^{\text{bn}}$
Intended: 'The child has been playing for two hours. / The child was playing for the last two hours.'

Furthermore, incompatibility with imperfective verb forms supports the proposal that BİT and 'bn' might be perfective signs. This is because perfective and imperfective are mutually exclusive, and the occurrence of the two forms together would yield ungrammaticality. Then, the next question is whether BİT and 'bn' nonetheless have a specific time reference. We are going to show that they have past time reference, and further that BİT can appear within a context that has a future reference, as would be predicted for a completive marker rather than a past tense marker.

Both BİT and 'bn' were found acceptable with past tense adverbials, as in (26a) and (26b), and present tense adverbials, as in (26c) and (26d). Even though they can occur with the present tense adverbial TODAY, the meaning in these sentences is recent past, as was also observed by Kubuş & Rathmann (2009). That is, the events 'study' and 'work' are perfective events. This shows that when the event includes either BİT or 'bn', its time reference is not in the present. This observation is in line with the observation that most languages have a strong tendency for perfective having past time reference (Dahl 1985).

10. Note that TİD has more than one strategy to mark imperfectivity, such as the mouth gesture 'lele' (Dikyuva 2011) and repetitive hand movement (Gökgöz 2011). All strategies were tested for this study.

(26) Context: Your parent asks you whether you studied for your class, and you tell him/her that you studied.

a. YESTERDAY STUDY BİT
'I studied yesterday.'

bn

b. YESTERDAY STUDY
'I studied yesterday.'

c. TODAY STUDY BİT
'I (have) studied today.'

bn

d. TODAY STUDY
'I (have) studied today.'

When BİT and 'bn' were presented with future time adverbials, the participants found (27a) and (27b) ungrammatical. Even though a perfective or an imperfective event theoretically can be represented in any time frame, the observations regarding (27a) and (27b) indeed follow the pattern in other languages. That is, the perfective has past time reference in some languages (Dahl 1985).

(27) Context: Your parent asks you whether you studied for your class and you tell him/her that you will study tomorrow, or you will finish studying tomorrow.

a. *TOMORROW STUDY BİT
'I will study tomorrow.'

bn

b. *TOMORROW STUDY
'I will study tomorrow.'

In terms of time adverbials, BİT and 'bn' have shown the same compatibility features. In other words, both are acceptable with past (26a)/(26b) and present tense (26c)/(26d) adverbials with past time reference, whereas they are unacceptable with future time adverbials (27a)/(27b). However, they behave differently in future contexts. Both sentences (28a) and (28b) were presented in the same context. The context is that you want to meet with your friend tonight, but your friend says that s/he has to study. You say that you can meet after s/he is finished studying, and that s/he should let you know. In this context, the event times of studying and informing are in the future even though no future tense adverbial is present. BİT and 'bn' were tested in this context, and it turns out that BİT is acceptable whereas 'bn' is not. When there is no overt future tense adverbial, but the time reference is future, BİT and 'bn' have distinct compatibilities. BİT is acceptable in the context that refers to completion in the future (when the studying is done), as seen in (28a) (Figure 7). In contrast, 'bn' is unacceptable with future time reference (28b).

(28) Context: You want to meet with your friend tonight, but your friend says that s/he has to study. You say that you can meet after s/he is finished studying, and that s/he should let you know.

a. STUDY BİT $_2$LET-ME-KNOW$_1$
'After you study/When you finish studying, let me know.'

bn

b. *STUDY BİT $_2$LET-ME-KNOW$_1$
'After you study/When you finish studying, let me know.'

SUBJECT STUDY BİT $_2$LET-ME-KNOW$_1$

'After you study/When you finish studying, let me know.'

Figure 7. Manual sign BİT in future context

To sum up, this section showed that BİT and 'bn' are not tense markers, as they are incompatible with imperfective verb forms. They are acceptable with past and present tense adverbials by conveying that the event has reached its end. Yet, neither is acceptable with overt future time adverbials. However, the judgments were different when BİT and 'bn' were presented in the same contexts where the event time is future (28). Even though there may be other factors such as mood causing the difference between (28a) and (28b), it is left for future studies why BİT and 'bn' behave differently in this context.[11]

11. While some languages have past time reference with perfective structures, others allow the perfective to have future time reference in specific constructions, like Arabic perfective in conditional sentences (Dahl 1985: 80). This may be the case that we observed with BİT in (28a) while 'bn' strictly has past time reference (28b). Another explanation for why they are not acceptable with future time adverbials can be mood distinctions. Future is treated as mood rather than tense in some languages where the distinction is between real and unreal, or realis and irrealis (Comrie 1985; Palmer 2001). If TİD treats future as a mood rather than tense, the unacceptability of (27b) and (28b) is expected. In other words, the future tense adverbial will convey irrealis, whereas the perfective form will entail the existence of the event. Thus, the combination of the existential and the irrealis will create a semantic mismatch. Regarding this, TİD may require a different morpheme to mark perfective in the future such as the Turkish evidential in future perfect structures (Arslan-Kechriotis 2006).

6. Negating BİT and 'bn'

Usually, the patterns of aspect markers when used in the context of negation are not the primary aim in the event semantics and aspect literature. However, an important observation that has been reported for various sign languages is the incompatibility of aspect morphemes with regular sentential negation (Fischer & Gough 1999; Meir 1999; Rathmann 2005; Zucchi et al. 2010). To show that TİD displays the same pattern as other sign languages, this section is devoted to checking the compatibility of the two aspect morphemes with the marker of sentential negation NOT (Figure 8). A sentence with either BİT or 'bn' is negated with HİÇ[12] (Figure 9) or the nonmanual marker 'ap'[13] (Figure 10), respectively. How these two aspectual morphemes are negated will also serve as a test to observe what happens when BİT and 'bn' co-occur in Appendix I.

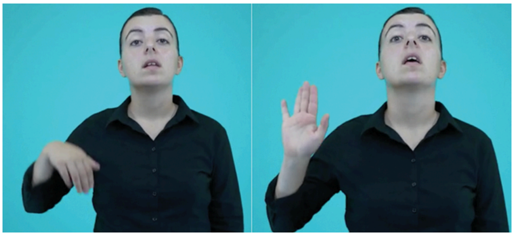


Figure 8. The marker of sentential negation NOT[14]

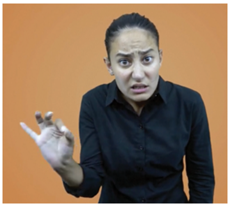


Figure 9. Negative manual sign HİÇ ('(n)ever')

Both BİT and 'bn' were separately presented to the participants with sentential negation NOT, as can be seen in (29). All sentences were judged as unacceptable.

12. HİÇ can be also signed two-handed.

13. The label 'ap' was first used in Dikyuva et al. (2017: 223–224) to refer to this specific nonmanual marker. Their convention will be used in this paper to have a consistent reference in the literature.

14. The images of the signs NOT and HİÇ are from tidsozluk.net.

Figure 10. Negative nonmanual marker 'ap' combined with the verb EAT

The unacceptability with BİT in (29) further shows that BİT is used as an aspect morpheme rather than a verb that takes an embedded clause.

(29) Context: You are in China but you haven't tasted the food there before. You're curious about the food.

a. #IX$_1$ EAT^NOT BİT

b. #IX$_1$ EAT BİT^NOT

bn

c. #IX$_1$ EAT NOT

bn

d. #IX$_1$ EAT^NOT

Intended Meaning: 'I did not eat (it). / I have not eaten (it).'

When the participants were asked to provide the acceptable forms of these sentences, they used the negative version of BİT, that is, HİÇ ('(n)ever') (30a) (Figure 11), or the mouth gesture 'ap' (puffing cheek(s), then releasing air) instead of the nonmanual 'bn' (30b).

(30) Context: You are in China but you haven't tasted the food there before. You're curious about the food.

hs

a. CHINA COME IX FOOD EAT HİÇ

'I haven't eaten Chinese food.'

ap

b. IX$_1$ EAT

'I didn't eat (it). / I haven't eaten (it).'

6.1 Negative HİÇ

Firstly, for HİÇ ('(n)ever'), Gökgöz (2011) noted that it can change the polarity of a sentence and does not need to co-occur with sentential negation NOT (31). Furthermore, Dikyuva et al. (2017) propose that HİÇ is the negative suppletive form of BİT. Dikyuva et al. (2017) note that when HİÇ appears in preverbal posi-

'I came to China and haven't eaten food there. (I haven't eaten Chinese food). I'm curious about it.'

Figure 11. Usage of HİÇ in a 'never' context

tion without being in the scope of NOT, the sentence is ungrammatical as in (32); that is, HİÇ needs to be in the scope of sentential negation (33). Additionally, HİÇ is observed after the sentential negation in Gökgöz's (2011) study, as seen in (34) below. As a brief interim summary, there are three patterns of use for HİÇ: (i) without a sentential negation and after the verb (31), (ii) before the verb with sentential negation (33), and (iii) after the sentential negation (34) (abbreviations in these examples are: ebr – eye brow raise, ht – head tilt, and nbp – non-neutral brow position).

(31) IX$_1$ TRY HİÇ (TİD, Dikyuva et al. 2017: 225)
'I have never tried (it).'

(32) *IX$_1$ HİÇ SEE (TİD, Dikyuva et al. 2017: 225)
Intended meaning: 'I have never seen (it).'

(33) ebr,ht (TİD, Dikyuva et al. 2017: 225)
IX$_1$ HİÇ SEE NOT
'I haven't ever seen (it).'

(34) nbp (TİD, Gökgöz, 2011: 69)
IX$_1$ SIGN KNOW NOT HİÇ
'I didn't know (how to) sign at all.'

Dikyuva et al.'s (2017) claim is that HİÇ is the negative perfective marker in those cases where it is acceptable without sentential negation. They further claim that HİÇ entails that the event has started but is not completed. In contrast, we argue that HİÇ entails the non-existence of the event. This is based on the fact that the participants preferred overt sentential negation for the cases in which the event actually is started but not completed, and they further commented that in order for sentences like (31) to be acceptable, the subject must not initiate the event.

Building upon the patterns observed here, we propose that HİÇ has at least two functions: (i) as seen in (33), it is a negative polarity item that is required to appear under NegP and be in the scope of the overt sentential negation; and (ii) as seen in (31), it may function as a negative aspect sign that follows the verb and scopes over it because it can negate the sentence without sentential negation, and thus patterns with negative aspect signs in other sign languages like NOT-YET in ASL, NON-ANCORA or NIENTE in LIS (Zucchi et al. 2010). The occurrence of HİÇ after sentential negation in (34) is explained based on a Spec-head relation and the spreading of the negative nonmanual onto HİÇ (Gökgöz 2011). One might consider HİÇ having the same function as an n-word (n-sign) in both cases and propose that sentences (33) and (34) display negative concord. This claim needs to be further investigated based on the behavior of negative signs in TİD.

6.2 'ap'

As for 'ap', Dikyuva et al. (2017: 223) claim that 'ap' is the negative counterpart of 'bn' because it indicates the noncompletion of the event. In contrast, as above for HİÇ, we argue that 'ap' entails the non-existence of the event rather than entailing noncompletion or termination.

The claim of noncompletion for 'ap' entails that the event may have started but may not yet be terminated. To test this claim, sentences marked by 'ap' were re-presented to the participants in contexts where the termination meaning is prominent. However, participants did not accept 'ap' in such contexts (35a), instead preferring to negate the sentence with the manual sign NOT (35b). Participants further noted that the verb form with the nonmanual sign 'ap' is acceptable only when the event has not yet started.

(35) Context: You have started to eat your sandwich, but you have not finished it. When your friend asks whether you finished eating your sandwich, you respond:

a. #IX$_1$ $\overline{\text{EAT}}^{\text{ap}}$
'I did not eat (it). (I have not finished eating it.)'

b. IX$_1$ EAT^NOT
'I did not eat (it).'

6.3 Summary of analyses of HİÇ and 'ap'

We have shown that both HİÇ and 'ap' entail the non-existence of the event as evidenced by the participants' evaluations of these examples. Crucially, we can argue that HİÇ and 'ap' do not only change the polarity of the sentence but also negate the existence of an event. Having different negative counterparts for BİT and 'bn' supports the proposal that they are different signs. Their difference will become clearer when we examine their aspectual features in Section 7.

7. Aspectual features of BİT and 'bn'

Up to this point, the tests for tense showed that BİT and 'bn' are not tense markers, and their differing negative counterparts support the suggestion that they behave like other perfective markers in sign languages. These findings also show that they are different morphemes. To further support this hypothesis, we tested both morphemes with different event types to understand their aspectual features. Before presenting the results of these tests in Sections 7.2 and 7.3, the definitions of perfectivity and telicity are reviewed in Section 7.1 to show that they are different semantic categories even though they have often been used interchangeably in the literature. The difference between perfectivity and telicity will help in explaining the difference between BİT and 'bn'.

7.1 Perfectivity and telicity

When the internal temporal structure of events is considered, two key notions need to be highlighted: perfectivity and telicity – notions that have sometimes been used interchangeably in the literature (for further discussion of this problem, see Bertinetto (2001)). Even though these semantic notions are universal, their realization in a specific language may differ. In other words, languages may weight tense and aspect differently in their grammar, so that they will have different realizations of these universal notions. For instance, some languages, such as English, may convey these notions via tense relations. As a result, it is difficult to attribute these notions to one specific grammatical form. Therefore, the possible interactions of telicity and perfectivity will be different in languages which express these notions in different grammatical forms, such as, e.g., Russian. Readers are referred to Bertinetto (2001) and Dahl (1985) for detailed discussion. To understand the semantic functions of BİT and 'bn', tests here are based on the universal notions rather than the coding in English or Turkish; perfectivity and telicity are treated as distinct semantic categories.

As for perfectivity, it reflects how an event is presented as either an open interval (36) (imperfective) or a closed interval (37) (perfective; cf. Bertinetto 2001; Comrie 1985; Smith 1997). These observations are defined in formal semantic terms as a potential containment relationship (Bhatt & Pancheva 2005). That is, if RT contains ET, it is imperfective (ET ⊂ RT) (36), whereas if ET is entirely before RT rather than contained in RT, then it is perfective (ET < RT) (37).

(36) The kid was painting a picture. [ET ⊂ RT]

(37) The kid painted a picture. [ET < RT]

In contrast, telicity is a different notion defined in terms of the event's temporal structure, which is often but not always lexicalized in a verb. Following Bertinetto's (2001) view that telicity is an inherent property of an event, we will refer to telicity when an *event* has an inherent endpoint independent of how it is lexicalized. That is, there may be telic events that require more than a single sign, just as multiple spoken words may be necessary. However, for this study, telicity will be considered only for the verb signs rather than verbal phrases or serial verb constructions because starting from the less complex structures will help to better understand the functions of BİT and 'bn'. This also allows us to use Ramchand's model (see below).

The crucial point is that both telic and atelic events can be presented in perfective and imperfective forms. Not all imperfective events entail that they are atelic; not all perfective events entail that they are telic, and vice versa (Bertinetto 2001). To illustrate, a telic event 'painting a picture' which has an internal endpoint (when the painting is finished) is presented with the imperfective in (36), where the event is ongoing and has not reached its endpoint. However, its presentation with the perfective is also grammatical in (37), where the event has reached its endpoint. Crucially, the event has an endpoint to be reached, making it a telic event regardless of the aspect in which it is presented in a given sentence.

Similarly, an atelic event can be represented with either imperfective or perfective. The tense of both the sentences in (38) and (39) is past since ET is before UT. The event 'run' is an atelic event which means that it does not have an inherent endpoint. Unless worldly restrictions are taken into consideration, the event can go on indefinitely. The event can be 'terminated' but not 'completed'.

(38) John was running for an hour.

(39) John ran for an hour.

Based on these universal semantic notions, it has been argued in the literature that states cannot be bound by a perfective marker whereas events can be (Bhatt & Pancheva 2005; Smith 1997; Zucchi et al. 2010, among others). Therefore, BİT and ‘bn’ were tested for their compatibility with different event types, as well as states and statives in order to determine whether BİT and ‘bn’ are perfective markers. As for states and statives, the distinction between them can be observed via individual-level predicates and stage-level predicates. Stage-level predicates like ‘being sick’ can change with time, so they should be acceptable with perfective morphemes. In contrast to stage-level predicates, individual-level predicates like ‘being tall’ do not change with time, and they should therefore not be compatible with perfective morphemes.

For event types, we use Ramchand’s event classification. She decomposes an event into initiation *initP*, process *procP*, and result *resP* (Ramchand 2008: 39–40). The combinations of [init], [proc], [res] and transitivity yield different classifications for lexical verb types. Table 2 represents the possible combinations of [init], [proc] and [res] in her categorizations (p. 108). The verbs in italics in the fourth column are the ones tested in this study.

Table 2. Ramchand’s event categorizations with the verbs tested in this study

	[*init, proc*]			**Telicity**
I	Transitive	INITIATOR, UNDERGOER	*drive, push, dance*	Atelic
	Transitive	INITIATOR, PATH	*eat, read, paint*	Telic
II	Intransitive	INITIATOR$_I$, UNDERGOER$_I$	*run, walk*	Atelic
	[*init, proc, res*]			
III	Transitive	INITIATOR, UNDERGOER$_I$, RESULTEE$_I$	*throw*	Telic
	Transitive	INITIATOR$_I$, UNDERGOER$_I$, RESULT-RHEME$_I$	*enter*	Telic
IV	Intransitive	INITIATOR$_I$, UNDERGOER$_I$, RESULTEE$_I$	*arrive, jump*	Telic
V	Ditransitive	INITIATOR, UNDERGOER, RESULTEE	*give, throw*	Telic
	[*proc*]			
VI	Intransitive	UNDERGOER	*melt, dry, freeze*	(A)telic[15]
	[*proc,res*]			
VII	Intransitive	UNDERGOER$_{I,}$ RESULTEE$_I$	*break, tear*	Telic

15. The verbs in this category are referred to as degree achievement in the literature. Languages may lexicalize these events either as a process (e.g., melting as an ongoing process from non-liquid to liquid – atelic), or as a process with an end point (e.g., melting as a process and the end result liquid – telic), or as only the endpoint of the event (e.g., completely melted, only liquid – telic) (Beavers 2012).

7.2 Aspectual properties of BİT

To analyze its aspectual features, BİT was further tested with Ramchand's (2008) different event categories and states. In terms of its aspectual features, the results indicate that BİT occurs with only [init, path] verbs and some [proc] verbs, and that it is unacceptable with all other categories. These patterns show that BİT is not a perfective morpheme that would be acceptable with all event types; instead, it picks specific events, namely those that involve a process and have a boundary. BİT is a completive morpheme that indicates that the event has reached its boundary.

BİT is found to be acceptable with [init, path] verbs like BUILD, PAINT or EAT, which have an endpoint. [init, path] is a subcategory of [init, proc] in Ramchand's categorization (as seen in Table 2). Since BİT makes a distinction between these subcategories of [init, proc], the results are reported separately here. Sentence (40) shows BİT following the event 'house building', in which BİT marks the endpoint. The interpretation is that the event has reached its right/end boundary, and the implication is the completion of the event unless it is clearly specified otherwise in the sentence or in the context.

(40) Context: You and your friend are chatting. Your friend asks you about the new house that they were building, and you respond:
IX-PL$_3$ HOUSE BUILD BİT
'They (have) built the house.'

Another crucial note from the discussion with the participants is that the implication of BİT is completion unless the sentences are modified with quantified noun phrases. BİT occurs with quantified noun phrases such as HALF in (41) or TEN PAGE in (42). The 'reading' event is mapped onto the theme 'the book'; the progress in reading goes along with the progress in the book. The quantified noun phrases in these sentences change the boundary of the event. Without quantification, the boundary is at the end of the entire event 'reading (whole) book' or 'eating (all) meat', whereas with quantification, the boundary is shifted by HALF and TEN PAGE. This is a one-to-one mapping of the event onto the theme and is called 'homomorphism' (Beavers 2012; Krifka 1987), which will be returned to in Section 7.1.2. Due to homomorphism, the whole event is understood as reaching the right boundary (the default target). Thus, HALF and TEN PAGE mark a different target boundary on the event, and BİT conveys that the event has occurred up to this target boundary. Rather than termination, BİT in (41) and (42) conveys that the event has reached its right boundary.

(41) Context: You go out to lunch with your labmates. You didn't finish your whole meal, and you say:
MEAT HALF EAT BİT
'I have eaten half the meat.'

(42) Context: Your classmate asks you whether you have read the assigned book for class:
BOOK TEN PAGE READ BİT
'I have read 10 pages.'

Ramchand's (2008) categories do not include states or statives; they only cover events. Nonetheless, we also tested states and statives with BİT and 'bn' to better assess their selectional properties. When BİT was presented with state verbs like KNOW, LIKE, and WANT, the participants found it unacceptable, as for example in (43). In terms of compatibility with states, its behavior was similar to what has been observed for FINISH in ASL and FATTO ('done') in LIS (Zucchi et al. 2010). This finding supports the notion that BİT has a specific aspectual function. Furthermore, incompatibility with states supports the proposal that BİT is not a tense marker because a state should be able to be presented in any time relation (Reichenbach 1947).

(43) Context: You went shopping with your kid and s/he asked for a toy.
#CHILD TOY WANT BİT
Intended meaning: 'The child wanted a toy.'

However, as shown in (44), BİT can occur with stative verbs like LIVE. Statives differ from states in terms of boundedness (Bertinetto 2001). That is, statives can be bounded by the perfective, as in (44), whereas states cannot (43). For the duration of the event (five years), there is no change in the state of living, but the situation is different at the end of the five years. In contrast to EAT and READ, LIVE does not involve a change during its duration. Yet, similar to them, LIVE also has a boundary when the location of living changes, and it can be used with BİT; it conveys that 'living in Istanbul' has reached its end as expressed by the following sentence 'She doesn't live (in Istanbul) now'.

(44) Context: You are talking about your friend Zeynep, and you know that she has lived in Istanbul.
ZEYNEP FIVE YEAR ISTANBUL LIVE BİT IX_3 NOW LIVE^NOT
'Zeynep (has) lived in Istanbul for five years, but she doesn't anymore.'

Beyond this, BİT is unacceptable with the remaining categories.[16] That is, it is incompatible with [init, proc, res] verbs like FALL, THROW, ENTER or GIVE, as shown for FALL in (45). BİT is also incompatible with verbs in the [proc, res] category, such as BREAK in (46). [res] is the crucial and shared property in both categories. Since these events already have a [res] in their lexical features, it is apparently unacceptably redundant for them to be marked for the endpoint.

(45) Context: There was a glass on the table, and the cat knocked it over while passing by.
#GLASS FALL BİT
Intended meaning: 'The glass fell. / The glass has fallen.'

(46) Context: There was a glass on the table. The cat knocked it over while passing by, and now it is broken.
#GLASS BREAK BİT
Intended meaning: 'The glass broke. / The glass has broken.'

On the other hand, the events 'run' in (47) and 'drive' in (48) involve a process but not an internal endpoint or change of state, and they are therefore classified as [init, proc] in Ramchand's representation. In terms of an eventive reading, BİT is also unacceptable with RUN and DRIVE even though both verbs involve a process. As stated before, [init, path] is considered a subcategory of [init, proc] in Ramchand (2008). Yet, BİT clearly makes a distinction between these subcategories, as it is acceptable with [init, path] but unacceptable with the remaining verbs in [init, proc]. The data, up to this point, shows that BİT selects events which have both a process and an endpoint.

(47) Context: You are talking about your daily activities, and you like running. You tell your friend that you ran this morning.
#IX_1 RUN BİT
Intended meaning: 'I have run/I ran.'

(48) Context: The signer is taking the driving test and his/her friend is, too. To check in with the signer, his/her friend asks whether s/he has taken the test or not. The signer replies:
#DRIVE BİT
Intended meaning: 'I have driven the car. / I drove the car.'

While BİT is consistently either acceptable or unacceptable with the verbs in a given category, its behavior with verbs in the [proc] category is inconsistent. That

16. Remember that these sentences are acceptable when BİT is a discourse marker or is sequencing the events, as shown in Section 4.1.

is, it is acceptable with some verbs like DRY (49), yet unacceptable with others like GROW (50). The main difference between DRY (49) and GROW (50) is a natural end of the event. In other words, the process of 'drying' will be completed once all the water has evaporated, while the process of 'growing' could continue in the domain of cognition even if physical changes may have stopped or are no longer observable. Although both events ('dry' and 'grow') have processes, their endpoints are treated differently with respect to their compatibility with BİT (49–50), and are in different verb classes at least in English (Levin 1993; indeed, the problem may be that 'grow' itself has additional constraints).

(49) Context: You did laundry today, and you hung the clothes to dry.
CLOTHES DRY BİT
'The clothes have dried.'

(50) Context: The signer's friend has not seen him/her for a while. When they see each other and chat, the friend asks about the signer's child.
#CHILD GROW BİT
Intended meaning: 'The child grew/ has grown.'

7.2.1 *BİT introducing [res] to the event*

As for the aspectual properties of BİT, it appears only with [init, path] verbs, statives, and some [proc] verbs, as can be seen in Table 3. Numbers in parentheses in the table refer to the sentences that were discussed in this section. When BİT appears with these events, it functions as a completive marker that indicates that the boundary has been reached. The implication is that stative (LIVE) continues, and that the events (BUILD and DRY) have not been completed if they are not marked with BİT. Furthermore, the boundary of the event can be moved by quantification such as HALF.

Another crucial point in Table 3 is that BİT does not appear with the categories that have [res] as part of the lexical item, which clearly indicates by itself that the event is completed. Instead, for events in those categories where completion might or might not have occurred, BİT conveys that the event has reached its endpoint. In view of crosslinguistic lexical variation, BİT can be considered to behave like some particles appearing with certain verbs in English. For instance, sentence (51) may mean that I ate my share and there's still some candy left. In contrast, 'eat up' in sentence (52) conveys that I ate all the candy, and there is no candy left. Ramchand analyzes particles like 'up' or 'in' as providing the [res] head for the event (2008: 131–138).

(51) I ate the candy.

(52) I ate up the candy.

Table 3. Compatibilities of BİT with Ramchand's categories (numbers in parentheses refer to TİD examples in this section)

Categories	Example	Compatibility	Telicity
States	WANT (43)	N	atelic
Statives	LIVE (44)	Y	(a)telic[17]
init, path	BUILD (40)	Y	telic
init, proc, res	FALL (45)	N	telic
proc, res	BREAK (46)	N	telic
init, proc	RUN (47)	N	atelic
proc	DRY (49)	Y	(a)telic
proc	GROW (50)	N	(a)telic

In line with her discussion, sentences without BİT are ambiguous; the implication is that the event reached its boundary, but this implication can be cancelled. Imagine (53) is the context for sentence (51). This context conveys that there is some candy left, and that the subject did not eat all the candy. In contrast, as seen in (54), sentence (52) cannot continue with 'but don't worry, I left some for you' because 'eat up' conveys that the event is completed and there is no candy left.

(53) I saw you had a candy bowl. I ate the M&Ms, which as you know are my favorite, but don't worry, I left some for you.

(54) I saw you had a candy bowl. I couldn't help myself, and I ate up the M&Ms, #but don't worry, I left some for you.

Similarly, BİT disambiguates the sentence and entails that the event is completed, just like 'up' does in (52). As shown in (54), trying to cancel the entailment in (52) yields a contradiction. A similar contradiction holds for the stative example

17. Statives can be argued to be either atelic or telic. They may be atelic because there is no internal change at a given internal point. For example, in sentence (i), there is no change in the event of 'standing' for an hour. In that sense, the event of 'standing' is like a state and can be argued to be atelic. In sentence (ii), 'resemble' behaves as a degree achievement like 'dry' or 'melt'. There is a change in resemblance but this change is not the ultimate end, and the change still continues. The degree achievement reading is also due to 'more and more' and not solely due to 'resemble' itself. Since statives can behave as either atelic (i) or telic (ii), they are considered as (a)telic (Bertinetto 2001).

(i) John deliberately stood in the way for an hour.

(ii) John is resembling his father more and more.

in (44), repeated in (55), which would not be acceptable if the sentence continued with 'and she still lives there'.

(55) Context: You are talking about your friend Zeynep, and you know that she has lived in Istanbul.
ZEYNEP FIVE YEAR ISTANBUL LIVE BİT IX$_3$ NOW LIVE^NOT
'Zeynep (has) lived in Istanbul for five years, but she doesn't anymore.'

Ramchand (2008) argues that the result(ative) meaning comes from the particles in these verbal phrases, while it comes from the homomorphic mapping of the event onto the theme in [init, path] events.[18] These events mostly behave like degree adjectives with a closed scale, and the event is completed when the theme is completed due to the closed scale. Briefly, the notion of homomorphism characterizes the one-to-one mapping between the unfolding of an event over time and its theme (Beavers 2012; Krifka 1987). The homomorphic relation is mostly found in verbs of creation like 'build', 'paint', or 'draw', in verbs of consumption like 'eat' or 'drink', or in degree achievements like 'melt' or 'freeze'. To illustrate, Figure 12 depicts the event of 'building a house'. As seen in the figure, the stages depict the parts of the event such as building the foundation, building the first floor, building the second floor, and so on. The event has stages, and the effects of these stages can be clearly observed on the theme itself. Progress on the event can be simultaneously observed on the theme. When the event is completed (i.e., finish building the house), the theme is also completed (i.e., the house exists). Likewise, with verbs of consumption, the theme has disappeared when eating or drinking is completed.

Figure 12. A depiction of homomorphic mapping between 'building' and 'house'

BİT is acceptable with verbs that potentially have homomorphic mapping like EAT, BUILD, WRITE, or (in a scalar sense) DRY. A puzzle for this account is pre-

18. See Ramchand's discussion on the difference between particles and prepositions. She argues that particles are the [res] head while prepositions are the complements to [proc] and the result meaning is derived differently (2008: 110–138). Since BİT and 'bn' do not behave uniformly with respect to the events in the [proc] category, we will treat BİT as a particle and leave the difference for future studies.

sented by statives like LIVE because there is no internal change in statives whereas gradual change is mapped onto the theme in other verbs. This is why we will treat BİT like the particles in English and leave the difference or similarities between degree verbs and statives for future research.

7.3 Aspectual properties of 'bn'

The current proposal that 'bn' is a perfective morpheme is supported by its occurrence with different event types but its ungrammaticality with states. Sentence (56) includes the state verb WANT with 'bn', which yields ungrammaticality, as expected. 'bn' was likewise tested with other state verbs such as STAND, KNOW, LOVE; its behavior was similar to what has been observed for FINISH in ASL and FATTO ('done') in LIS (Zucchi et al. 2010).

(56) Context: You went shopping with your kid, and s/he asked for a toy.

$\overline{}^{\text{bn}}$
*CHILD TOY WANT
Intended meaning: 'The child wanted a toy.'

On the other hand, 'bn' can grammatically occur with verbs like LIVE, as in (57) (compare with (55) which has BİT). Verbs like LIVE are statives and differ from states in terms of boundedness (Bertinetto 2001). For the duration of five years, there is no change in living, and the subject is in that state. Furthermore, the subsequent clause in (57) supports the notion that 'bn' is a perfective morpheme since it indicates that the state of living in Istanbul does not hold for the individual anymore and that 'live' is bounded.

(57) Context: You are talking about your friend Zeynep, and you know that she has lived in Istanbul.

ZEYNEP FIVE YEAR ISTANBUL $\overline{\text{LIVE}}^{\text{bn}}$ IX_3 NOW LIVE NOT
'Zeynep (has) lived in Istanbul for five years, but she doesn't anymore.'

The marker 'bn' was examined with different Ramchand event categories. As expected if 'bn' is a perfective marker, their occurrence with 'bn' is acceptable. Sentence (58) involves the verb BUILD from the [init, path] category. The event 'build' has an internal endpoint that 'bn' is compatible with. The implication is that the event has reached its endpoint before the reference time (RT) of the utterance.

(58) Context: You and your friend are chatting. Your friend asks you about the new house that someone was building, and you respond:

bn

IX$_3$ HOUSE BUILD

'They have built the house.'

[path] in the event can be quantified, as shown in (59) and (60). Similar to BİT, 'bn' is acceptable with these quantified themes which bound the event and entail that the event has occurred up to the target boundary of the quantified theme. Actually, the event can be considered as 'eating half of the meat' in (59) or 'reading 10 pages' in (60) (Beavers 2012; Krifka 1987), so the events in (59) and (60) have ended.

(59) Context: You go out lunch with your labmates. You didn't finish all your food, and you say:

bn

MEAT HALF EAT

'I ate half the meat.'

(60) Context: Your classmate asks you whether you have read the assigned book for class:

bn

BOOK TEN PAGE READ

'I (have) read 10 pages.'

To distinguish completion (reaching an endpoint) from termination, we will address the termination reading with the category [init, proc]. Termination of an event is relevant when an event does not have an endpoint, in other words, it is an ongoing process. Both states and the verbs in the [init, proc] and [proc] categories do not have an endpoint; however, they differ in terms of boundedness. States cannot be completed nor terminated, whereas the events in the [init, proc] category can be terminated. The difference is that the subject of a state is the undergoer and does not have control over the state, whereas the subject of [init, proc] is an agent and can stop the event. Thus, verbs in the [init, proc] category can be bounded with a perfective marker. Dikyuva's (2011) proposal that 'bn' is a completive marker predicts that 'bn' should not be acceptable with [init, proc] verbs, thus some other explanation is needed for the acceptability patterns shown by the examples below.

To test the occurrence of 'bn' with [init, proc] verbs, sentences (61) and (62) were presented to the participants. Both sentences include [init, proc] verbs, viz. RUN and DRIVE, which lack a [res], that is, an endpoint, as part of the lexical verb. These sentences show that 'bn' can bound a process and yield a termination meaning due to the compositional structure of the event. They, furthermore, confirm that ‘bn’ has more than just a completion reading, contrary to Dikyuva’s proposal.

(61) Context: You are talking about your daily activities, and you like running. You tell your friend that you have run this morning.

$\overline{}^{\text{bn}}$
IX$_1$ RUN
‘I have run. / I ran.’

(62) Context: The signer is taking the driving test and his/her friend is, too. To check in with the signer, his/her friend asks whether s/he has taken the test or not. The signer replies:

$\overline{}^{\text{bn}}$
IX$_1$ DRIVE
‘I drove the car. / I have driven the car.’

The marker ‘bn’ was also tested with verbs which contain both [proc] and [res]: [init, proc, res] in (63) and (64), and [proc, res] in (65). These verbs are considered ‘instantaneous’; essentially the events they express are completed as soon as they are started. As seen in sentences (63) to (65), ‘bn’ is indeed acceptable with verbs from these categories.

(63) Context: There was a glass on the table. The cat knocked it over while passing by.

$\overline{}^{\text{bn}}$
GLASS FALL
‘The glass fell. / The glass has fallen.’

(64) Context: There was a ‘for sale’ sign on the window of a house. You are passing by the area, and the sign is gone. You think:

$\overline{}^{\text{bn}}$
IX-PL$_3$ HOUSE SELL
‘They (have) sold the house.’

(65) Context: There was a glass on the table. The cat knocked it over while passing by, and it is broken.

$\overline{}^{\text{bn}}$
GLASS BREAK
‘The glass broke. / The glass has been broken.’

There is a further problem. Sentence (66) also involves an [init, proc, res] event, with the verb UNDERSTAND, yet ‘bn’ is unacceptable with this verb. When sentences (63), (64) and (65) are considered together, they differ in the role of their subjects. The subject in (64) is an agent. In (63) and (65), the subjects are undergoers, but an external or internal cause may be needed for the event ‘fall’ or ‘break’ to be realized. Like (63) and (65), in (66), the subject is an undergoer, and the event entails a change of state. But in contrast, while causation may be involved for FALL or BREAK, there is no causation or agentivity needed in UNDERSTAND. Thus, to be grammatical, 'bn' may require the event to have an agent or a cause.

(66) Context: The deaf consultant gives you instructions on the task, and you say that you understood them.

      bn
*IX$_1$ UNDERSTAND
Intended meaning: 'I (have) understood.'

Unacceptability with 'bn' is not unique to UNDERSTAND. It is also unacceptable with some verbs in the category [proc], as illustrated in (67) (Figure 13) and (68). This unacceptability is crucial with regard to the analysis because, theoretically, a perfective sign should be able to bound all eventualities, but 'bn' turns out to have different patterns with different eventualities.

(67) Context: The signer's friend has not seen him/her for a while. When they see each other and chat, the friend asks about the signer's child.

      bn
*CHILD GROW
Intended meaning: 'The child grew up. / The child has grown up.'

bn
GROW BE NOT

bn
'GROW is not acceptable.'

Figure 13. The participant's judgment on the acceptability of combining GROW with 'bn', as in (67)

(68) Context: Your kid is playing outside, and you ask him/her to come home because the sky is getting dark.

      bn
*SKY DARK
Intended meaning: 'The sky (has) darkened.'

The data, up to this point, consistently supports the hypothesis that 'bn' is a perfective sign, as it displays the same patterns reported in the literature for other sign languages (Meir 1999; Zucchi et al. 2010). However, 'bn' may require even more features from an event. What is the difference between the acceptable and unacceptable sentences? As in the sentence with UNDERSTAND in (66), all external arguments in (67) and (68) are undergoers. What would happen if the external arguments were replaced with agents, as in the example with SELL in (64)? 'bn' was also presented with [proc] verbs whose external arguments are agents and internal arguments are undergoers, (69) and (70). Both were found to be acceptable with 'bn', in contrast to (67) and (68). Even though the unacceptability of these latter two examples unifies them with (66), the question remains why 'bn' is acceptable with FALL in (63), which also has an undergoer subject, but not with those in (67) and (68)? We will return to this question in the following section.

(69) Context: Your labmate requested you to open the window and you reply:

IX$_1$ WINDOW $\overline{\text{OPEN}}^{\text{bn}}$

'I (have) opened the window.'

(70) Context: Your labmate requested you to close the window and you reply:

IX$_1$ WINDOW $\overline{\text{CLOSE}}^{\text{bn}}$

'I (have) closed the window.'

To conclude this section, 'bn' makes a distinction between states and events; 'bn' bounds the events. In terms of containment relations (Bhatt & Pancheva 2005), 'bn' encodes that the event took place before the reference time [ET < RT]. Based on this relation, we conclude that 'bn' is the perfective morpheme, which is located in AspP in terms of hierarchical relations. This proposal is supported by its incompatibility with states and its compatibility with all categories in Ramchand (2008) with a few exceptions (Table 4). We have thus provided a more comprehensive analysis for 'bn' than previous studies (Kubuş & Rathmann 2009; Dikyuva 2011; Dikyuva et al. 2017).

7.4 Does 'subject' play a role in the acceptability of BİT and 'bn'?

As stated in the previous section, 'bn' has different acceptabilities with verbs in the [init, proc, res] and [proc] categories like UNDERSTAND, GROW, or MELT. One may wonder whether the agentivity of the subject changes its acceptability because all external arguments in (66), (67) and (68) are undergoers. MELT has been argued to be a labile verb in terms of the causative-inchoative alternation (Kayabaşı 2020). Kayabaşı observed that the lexical verb alone was used for the inchoative

Table 4. Compatibilities of 'bn' with Ramchand's categories (numbers in parentheses refer to TİD sentences in this section)

Categories	Example	Compatibility	Telicity
States	WANT (56)	N	atelic
statives	LIVE (57)	Y	(a)telic
init, path	BUILD (58)	Y	telic
init, proc, res	FALL (63)	Y	telic
proc, res	BREAK (65)	Y	telic
init, proc	RUN (61)	Y	atelic
Proc	OPEN (69)	Y	(a)telic
Proc	GROW (67)	N	(a)telic

meaning (71), whereas a handling classifier predicate HDCL-PAN is added along with the verb for causative meaning (72). A third strategy for causatives reported by Kayabaşı (2020) resembles serial verb constructions. In our data collection, when asking the participants to discuss MELT, we observed a similar pattern with the verb PUT signed before MELT (73). Sentence (73) requires further testing to see if it involves coordinated clauses or a serial verb construction; however, the participants, while allowing 'bn' on PUT, did not accept 'bn' with MELT in (73).

(71) ICECREAM MELT (Kayabaşı 2020)
'The ice cream melted.'

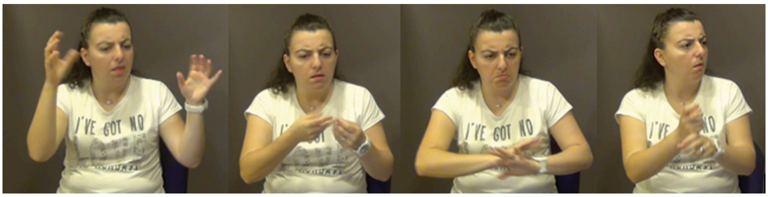
(72) HDCL-PAN BUTTER PUT, WOMAN HDCL-PAN MELT (Kayabaşı 2020)
'The woman melted the butter'

(73) Context: You're giving a recipe to your friend:
bn
BUTTER PUT MELT
'Melt the butter.'

BİT also showed variety in terms of acceptability with MELT. Strikingly, the acceptability changes with the subject. Combination of BİT with MELT was found unacceptable with BUTTER (74) but acceptable with SNOW (75). BİT might require the theme to reach its boundary in addition to the event reaching its endpoint. In other words, butter changes its form when melted, but there is no snow left once it melted – as indicated by the sign EMPTY in (75) (Figure 14).

(74) #BUTTER MELT BİT
'The butter melted.'

(75) Context: I was talking with mom, and I know that it snowed. We facetimed and the snow melted.
SNOW MELT BİT EMPTY
'The snow melted.'



SNOW MELT BİT EMPTY

'The snow melted.'

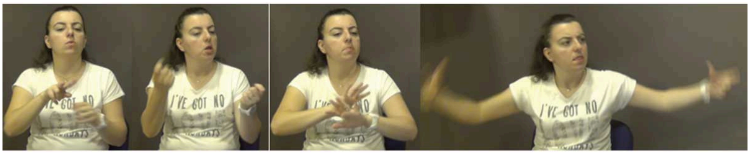
Figure 14. Acceptability of BİT with SNOW MELT

'bn' has been observed to display similar variation depending on the subject. As seen in sentence (76), the predicate ENLARGE was tested with the subject ROAD, and the participant found 'bn' over ENLARGE unacceptable by commenting that she has never seen it signed that way (Figure 15). However, during the discussion, the participant provided sentence (77) and commented that it is acceptable (Figure 16). That is, similar to the BUTTER MELT and SNOW MELT examples above, 'bn' also showed different acceptabilities with different subjects (ROAD vs. BRAIN) for the same verb (ENLARGE).

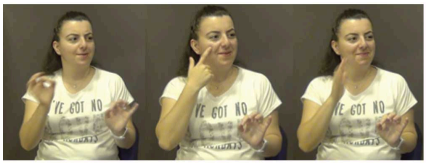
(76) Context: There was road construction to enlarge it. The construction has ended, and they enlarged the road.
bn
*ROAD CONSTRUCTION BİT ENLARGE
'The road construction is completed; the road is enlarged.'

(77) Context: During the discussion of sentence (76), the participant provided the following example:
bn
CHILD BRAIN ENLARGE
'The child's brain developed. / The child cognitively developed.'

Based on these few but crucial examples, the nature of the subject seems to play a role with [proc] verbs. It may be the case that either agentivity, verb type, or their combination affects the acceptability patterns. These constructions and their effects on BİT and 'bn' need to be further analyzed in future studies.



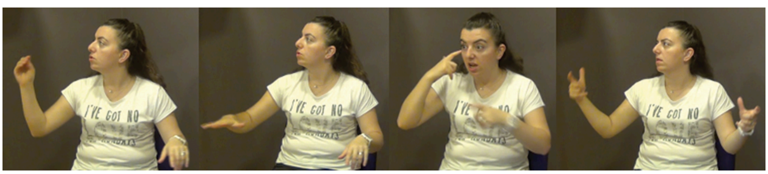
ROAD CONSTRUCTION BİT ENLARGE (bn)



HİÇ SEE NOT

'I have never seen ENLARGE.' (bn over ENLARGE)

Figure 15. The participant's judgment on the acceptability of combining ENLARGE with 'bn', as in (76)



CHILD BRAIN ENLARGE (bn)

'The child's brain developed. / The child cognitively developed.'

Figure 16. 'bn' is acceptable with BRAIN ENLARGE (bn over ENLARGE)

8. Conclusion

The central aim here was to define the aspectual properties of BİT and 'bn'. Their semantic properties show that they are separate signs based on their temporal features and the contexts that they appear in. First, based on the contexts where BİT occurs, we conclude that it is a multi-functional morpheme similar to FINISH in ASL (Fischer & Gough 1999). In contrast, 'bn' appears to have the same function (perfective morpheme) even when it appears in past, perfect or impersonal sentences.

The difference between BİT and 'bn' becomes clearer when their aspectual properties are examined. Their compatibilities with different event types clearly revealed: 'bn' occurs with all event types and statives except states as seen in Table 5, and as a result, we have argued that it is a perfective morpheme hosted by AspP. Based on the event types that it co-occurs with, 'bn' yields either a termination or completion reading. More precisely, if it bounds an atelic event, it yields the termination reading, and if it bounds a telic event, it yields the completion reading. In this respect, 'bn' creates a morphologically complex sign by bounding the event.

Table 5. Compatibilities of BİT and 'bn' with Ramchand's categories (numbers in parentheses refer to TİD sentences in the related sections)

Categories	Example	BİT	'bn'	Telicity
States	WANT	N (43)	N (56)	atelic
Statives	LIVE	Y (44)	Y (57)	(a)telic
init, path	BUILD	Y (40)	Y (58)	telic
init, proc, res	FALL	N (45)	Y (63)	telic
proc, res	BREAK	N (46)	Y (65)	telic
init, proc	RUN	N (47)	Y (61)	atelic
proc[19]	DRY	Y (49)	–	(a)telic
proc	OPEN	–	Y (69)	(a)telic
proc	GROW	N (50)	N (67)	(a)telic

On the other hand, BİT appears with statives, [init, path] verbs, and some [proc] verbs, and entails that the event has reached its right boundary. As seen in Table 5, all the acceptable events involve a process and a boundary. When BİT occurs with these events, it introduces a [res] to the event. This entails that the event reached its target final point even if that endpoint is modified by a quantified phrase, which moves the target boundary. The quantification changes the location of the boundary, and BİT marks that the event occurred until it reached that target boundary.

Regarding the pragmatic role of BİT in context, it is left for future studies to explore whether all the functions described here are related to one core semantic function, and whether it plays a role at the 'at-issue' level to mark the completedness of events like 'run' or 'drive', which are usually considered as ongoing. In addi-

19. Since this study is the extended version of an earlier study, the data is re-examined in terms of Ramchand's (2008) framework. The previous data did not have the same verb in the [proc] category for BİT and 'bn', yet different [proc] verbs could still be used in the analysis of BİT and 'bn'.

tion to BİT, the analysis of HİÇ would benefit from follow-up studies regarding the relation between the one behaving as the negative aspect sign and the one behaving as the negative polarity item. Last but not least, future study should include the cases where BİT and 'bn' co-occur; we provide examples in Appendix I. In addition to understanding their semantic relations, these cases would also confirm the hypothesized hierarchical relationship between them. For now, we conclude that 'bn' is higher, as it sits in the Asp head as a perfective morpheme, than BİT, which occupies the lower [res] head.

Aspectual properties of BİT and 'bn' support the conclusion that they are different morphemes even though they can co-occur. In terms of aspectual properties, the manual sign BİT is not similar to manual perfective signs in other sign languages, like FINISH in ASL or FATTO in LIS. FINISH and FATTO occur with all event types (Zucchi et al. 2010) while BİT has more restricted selectional properties. Unlike BİT, 'bn' is the one that occurs with all event types except states and is the perfective marker. In other words, TİD features a nonmanual marker 'bn' for a perfective morpheme, in contrast to other sign languages, where manual signs are used for the same purpose. The properties of BİT and 'bn' make TİD typologically different from other sign languages studied so far.

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Appendix I. When BİT and 'bn' co-occur

As briefly mentioned in Section 1, BİT and 'bn' can occur together, see (2b), repeated here as (78). The previous sections on temporal features and negation have also demonstrated that the two morphemes show different properties with some structures. Then, the following question arises: What happens when they co-occur? Two environments can show how they behave together: future contexts and negation.

(78) Context: The signer was instructed to imagine that she is giving a TİD class and that she was going to start a new topic, and to tell the class that she explained/ covered ('told') the previous chapter.
bn
IX$_1$ TELL BİT
'I (have) told (it).'

Future context

The first relevant environment is with future time reference: BİT is compatible while 'bn' is not. If we recall sentence (28a), BİT can have future time reference when it sequences events. On the other hand, as was seen in (28b), 'bn' is not acceptable in the same sentence. Both examples are repeated as sentences (79a) and (79b), respectively. The participants were presented with BİT and 'bn' co-occurring in the same sentence with future time reference (79c). As the unacceptability of (79b) suggests, the restriction on the use of 'bn' alone blocks (79c).

(79) Context: You want to meet with your friend tonight, and your friend says that s/he has to study. You say that you can meet after s/he is finished studying, and that she should let you know.

a. STUDY BİT $_2$LET-ME-KNOW$_1$
'After you study/When you finish studying, let me know.'

bn
b. #STUDY $_2$LET-ME-KNOW$_1$
Intended meaning: 'After you study/When you finish studying, let me know.'

bn
c. #STUDY BİT $_2$LET-ME-KNOW$_1$
Intended meaning: 'After you study/When you finish studying, let me know.'

Negation

Negation is the second relevant environment to compare the behavior of BİT and 'bn'. Two different negative markers, viz. HİÇ ('ever') and 'ap', were observed in place of BİT and 'bn' under negation (Karabüklü 2016; Dikyuva et al. 2017). Our previous Examples (30a) and (30b) are repeated in (80b) and (80c). The participants were given sentences with both BİT and 'bn', like the one in (80a), and were asked to negate these sentences. They preferred to negate such structures with 'ap' (80b) rather than with HİÇ (80c).

(80) Context: Your friend asks you whether you have eaten Chinese food.

bn
a. IX$_1$ EAT BİT
'I ate (it). / I have eaten (it).'

ap
b. IX$_1$ EAT
'I didn't eat (it). / I haven't eaten (it).'

c. IX$_1$ EAT HİÇ
'I didn't eat (it). / I haven't eaten (it).'

In sum, both future context and negation indicate that 'bn' preserves its features when BİT and 'bn' co-occur. That is, due to 'bn', the structure with both morphemes present is unacceptable in the future context. Also, the sentence with both morphemes is negated with 'ap', which is the negative suppletive form of 'bn'. This data may also show their hierarchal positions; however, this will be left to future studies.

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